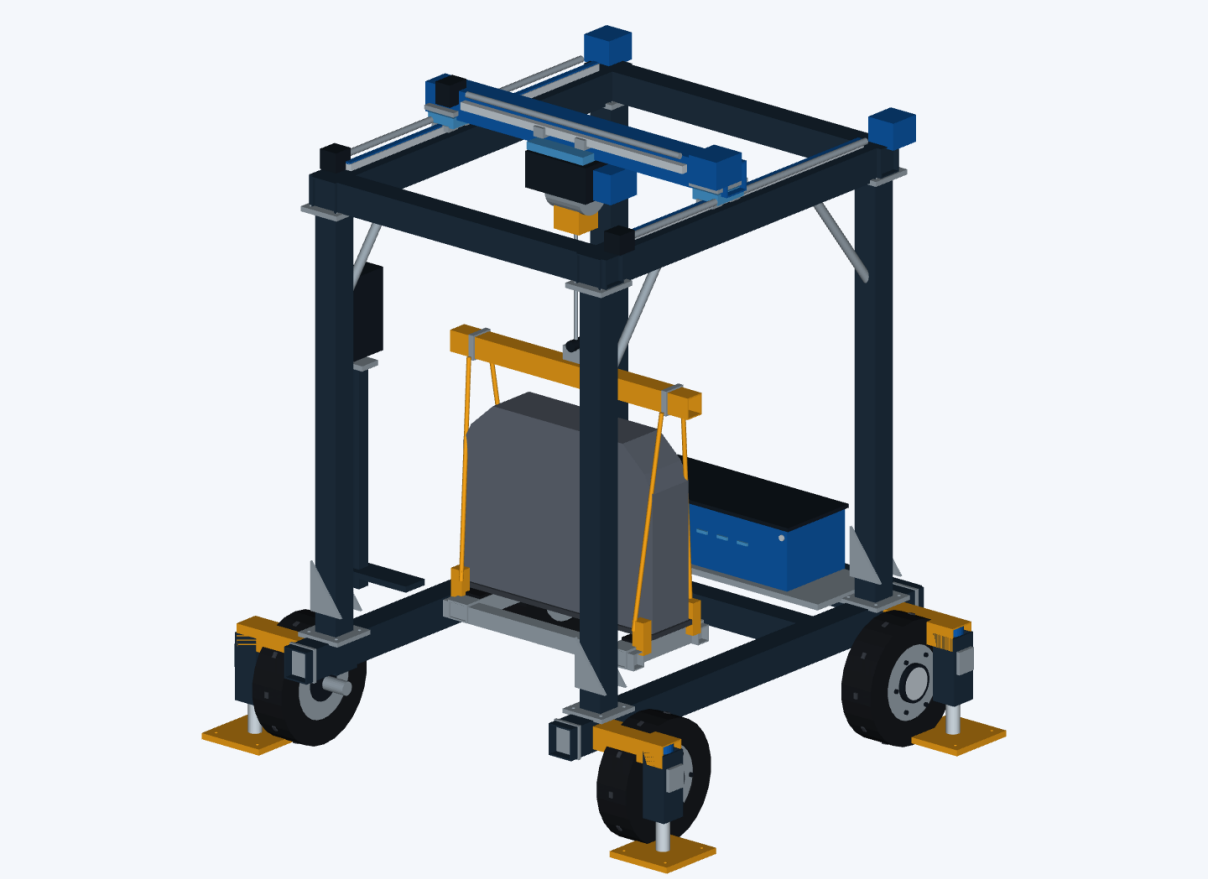
TECHNICAL REPORT / OCTOBER 2026

RoverSetter

Mechanical design and longitudinal stability investigation



RoverSetter CAD assembly, current export.

Summary

RoverSetter began as a response to difficult cemetery monument handling. I built three physical prototype iterations and observed longitudinal instability during a small suspended-stone test with Prototype 3. I then modeled how moving a suspended load forward affects the rear wheel reaction. The software reproduces static calculations for defined hypothetical inputs; the original motor-startup response remains an open dynamics question.

Technical scope: rigid-body statics and documentation of prototype observations. The 2,000 lb design goal was not achieved or load-tested.

01 / PROJECT DEVELOPMENT

From field problem to prototype

Installation problem

An installation through my monument business required approximately three to four hours, and the stone was damaged during handling. The experience motivated a compact, maneuverable machine concept for transportation and precise placement in cemetery conditions.

Design development

Prototype	Work completed	Engineering outcome
1	Initial mechanical proof of concept	Established the first arrangement.
2	Revised mechanical layout	Continued work on the handling mechanism.
3	Motorized chassis and forward hoist/positioning concept	Exposed a longitudinal stability issue during testing.

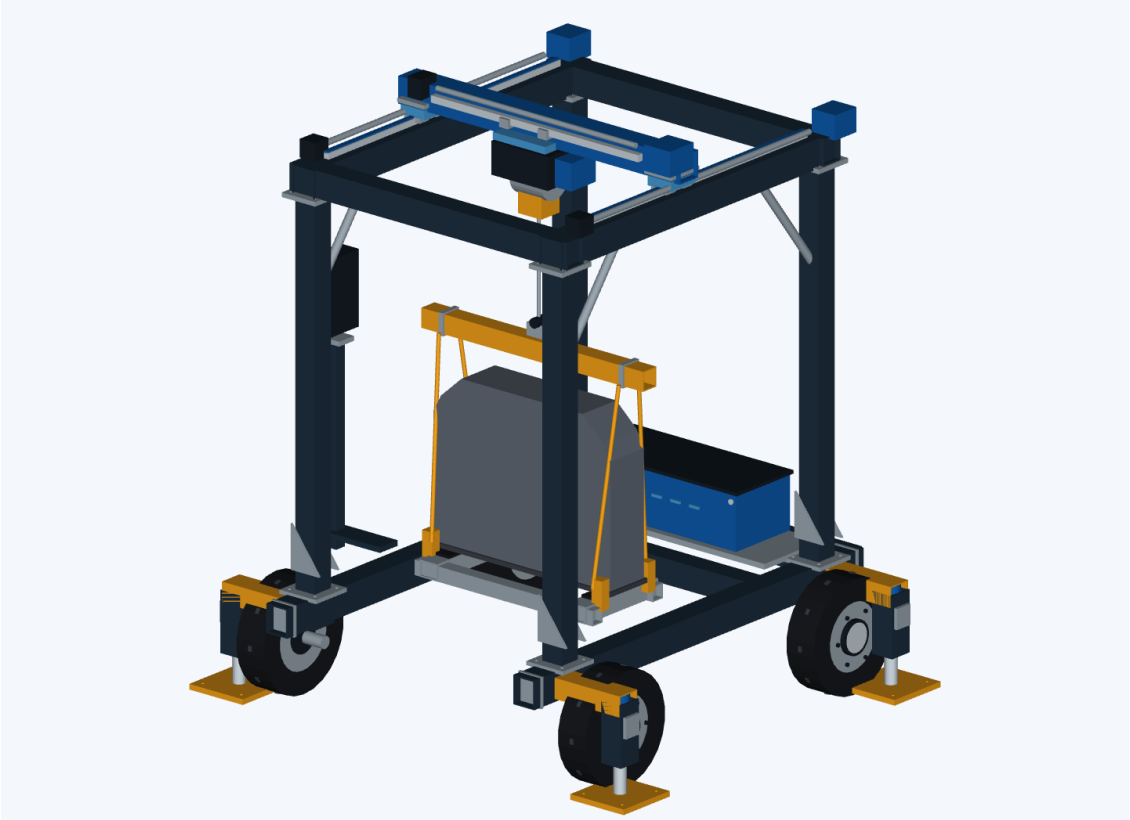
Current engineering record

The original machines no longer exist. Current CAD models are maintained exports of the RoverSetter design. A complete component-level record of the first two builds and their changes is unavailable; the third prototype has the clearest retained test account.

Design target

The project aimed to develop equipment capable of handling up to 2,000 lb while remaining compact enough for cemetery access. The numerical study in this report does not assign a structural or load rating to the machine.

Machine architecture



Main arrangement: mobile lower frame, upright support structure, top positioning carriage, wheel assemblies, and represented stabilizers.

Available CAD geometry

Design feature	CAD parameter
Chassis plan	1,360 × 1,950 mm
Wheel-center spacing	1,536 mm transverse; 1,290 mm fore-aft
Nominal wheel diameter	450 mm
Gantry top elevation	2,050 mm

These are geometry parameters represented in the current design files. The original physical machine’s full as-built weight distribution and contact forces were not independently measured.

03 / HISTORICAL TEST

Prototype 3 stability observation



Frame-and-stone CAD view. The displayed stone placement does not define the test overhang.

Test conditions

Condition	Recorded account
Surface	Hard ground
Support	Chassis supported on its wheels
Stone	Approximately 70 lb; about 16 × 8 × 6 in
Load position	Suspended beyond the front support line
Motion	Stationary first; powered chassis acceleration afterward

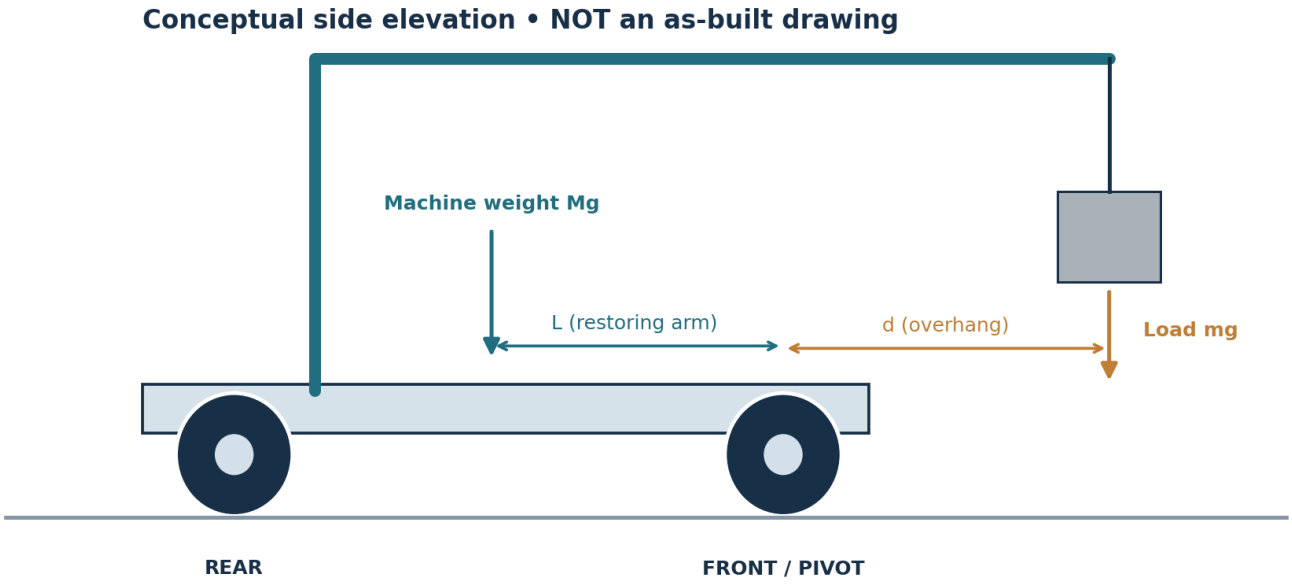
Sequence

While stationary, both rear wheels rose nearly together, followed by rocking and settling. When the drive motors started moving the chassis forward, pitching became more pronounced.

Measurements not available

The actual hook overhang, chassis mass and center of gravity, individual wheel forces, motor torque, and time history of chassis and stone motion were not retained. The observation is therefore a test record, not a quantified failure threshold.

Longitudinal static equilibrium



Simplified gravity loads and front-pivot moment arms.

Moment balance

Take moments about the front wheel-contact line. The machine weight produces a restoring moment MgL, while the load weight creates an opposing moment mgd. Rear support acts through the wheelbase distance B.

$$N_{rear} = (M g L - m g d) / B$$

Sensitivity to overhang

Keeping all masses and geometry fixed, differentiating with respect to forward overhang gives:

$$dN_{rear} / dd = -m g / B$$

The negative derivative means that a larger forward overhang reduces the rear support reaction predicted by this model.

Model assumptions

The machine is stationary, the ground is level, the chassis is treated as rigid, and load forces are vertical. This model groups the rear wheels into a single equivalent support and cannot resolve left/right wheel differences or motor-startup dynamics.

Reproducible calculations

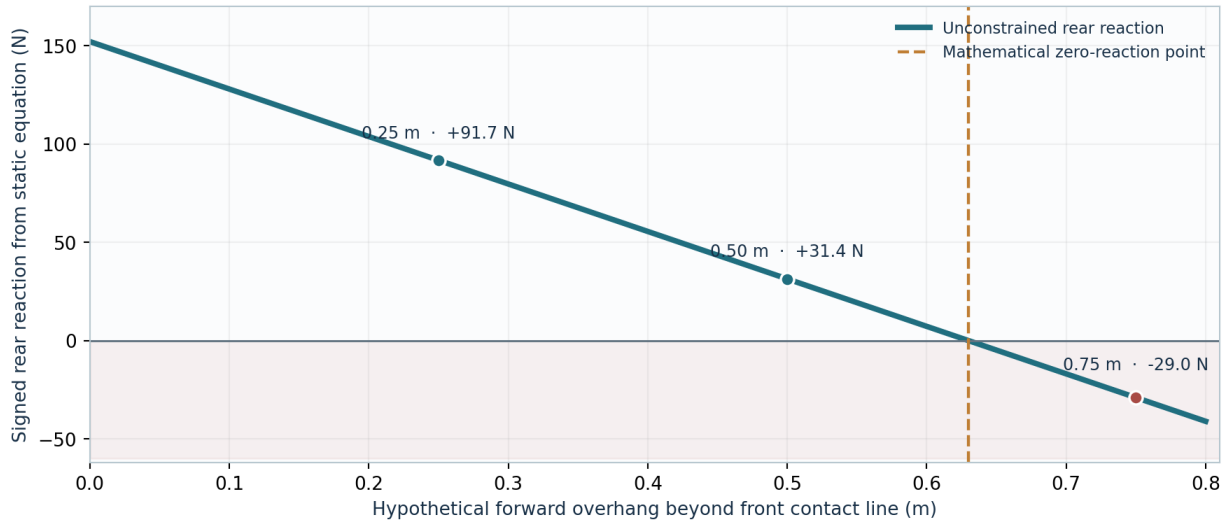
Reference parameters

Parameter	Value	Basis
Machine mass M	40 kg	Hypothetical
COM restoring arm L	0.50 m	Hypothetical
Wheel spacing B	1.29 m	CAD design value
Load mass m	≈31.75 kg	Approx. 70 lb recollection
Forward overhang d	0.25 / 0.50 / 0.75 m	Hypothetical

Calculated results

d	Rear reaction	Interpretation
0.25 m	+91.7 N	Positive rear reaction
0.50 m	+31.4 N	Smaller rear reaction
0.75 m	−29.0 N	Four-contact solution infeasible

Forward overhang lowers theoretical rear support

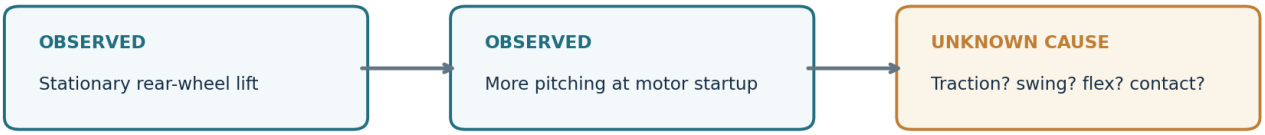


MODEL ONLY: M=40 kg, L=0.50 m, B=1.29 m (assumed); m≈31.75 kg (recalled estimate). Negative values are INFEASIBLE contacts.

The signed negative output is a mathematical indication that the assumed wheel-contact state cannot be sustained, rather than a real negative wheel force.

Why motor startup remains unresolved

The startup observation is evidence; its cause is not yet identified



Two separate mechanisms

Positioning carriage: moving a load forward relative to a stationary chassis changes the gravitational moment arms. The previous static model addresses this effect.

Motorized chassis: accelerating the complete machine introduces inertial response, drivetrain reaction, possible suspended-load swing, structural compliance, and changes in wheel contact. Static moment balance alone does not determine their combined effect.

What the observation establishes

I observed stronger pitching immediately after powered motion began. The original motion and load histories are not measured, so I cannot isolate an individual cause from that sequence.

Future model development

A dynamic investigation would need synchronized measurements of chassis acceleration, body pitch, wheel reactions, load motion, and motor command/torque. The static analysis is a foundation for that work, not its replacement.

Engineering conclusions

Established from the model

Under the stated assumptions, a suspended load moving farther beyond the front support decreases the calculated rear-wheel reaction. The sensitivity is linear in overhang at fixed load mass and wheel spacing.

Observed during the physical test

Prototype 3 exhibited near-symmetric rear-wheel lift and rocking while stationary, then stronger forward pitching at drive startup. The historical test did not yield calibrated wheel-force or time-resolved motion data.

Software verification

The Python model and seven unit tests run through GitHub Actions. The checks cover arithmetic balance, input handling, reaction slope, reference scenarios, and moving-carriage sensitivity.

Next engineering work

The next appropriate step is a teacher-reviewed tabletop mechanics experiment using a light, securely attached mass on a stable stationary apparatus. Measurements of support reaction versus mass position could be compared with the predicted slope without replicating a hazardous heavy-load event.

Project boundary

The original 2,000 lb goal remains a target. No rated lifting machine, experimentally measured historical tipping threshold, or definitive startup-failure diagnosis follows from the current data.

Associated material: two-page case study, design CAD views, Python source, tests, and reference scenario CSV.